

A LIKELY STORY PLATO'S *TIMAEUS*

Imagine, if you will, that you're a medieval monk in a well-stocked library. It's the eleventh century AD. One day you go to the shelf where Plato's writings are kept, and find there a single volume. You open it, and begin to read. You aren't going to be reading the *Republic*, because it hasn't been translated into Latin. Nor will you be reading the *Phaedo* or the *Meno*, neither of which will be available in Latin until the middle of the twelfth century. The Platonic corpus has fortunately been preserved and studied in the Greek-speaking medieval Byzantine empire. Its rediscovery in Western Europe, where Latin is still the language of scholars, will help spark the Renaissance. But that's still centuries away. So what you hold in your hands represents the complete works of Plato as they are known in your time and place: an incomplete Latin translation of a single dialogue, the *Timaeus*. Although Plato is hardly more than a name to you, the themes of this work will interest you greatly as you leaf through its pages. For it is in the *Timaeus* that Plato presents his thoughts on the creation of the world and the providential order of the universe—topics that are close to your heart, since you are a medieval monk. In the *Timaeus*, you discover what Plato has to say about god.

Plato's god is, however, rather different from the God of medieval monks. We can tell this already from the way Plato refers to him. Although it is made clear that we are dealing with a god (30a), Plato also calls him a craftsman—in Greek, *demiourgos*. This so-called “Demiurge” has two things in common with human craftsmen which make him unlike God as He is usually understood in Christianity, Judaism, and Islam. Firstly, the Demiurge is working from a set of plans. Just as a house-builder might follow architectural drawings, so the Demiurge looks to the Forms. He seeks to create an imitation of these Forms, and this imitation is the physical universe. Secondly, the Demiurge does not create from absolute nothingness. Instead, he fashions the universe in a kind of receptacle, without which he would have nowhere to put anything. Thus the Demiurge, unlike the God of the revealed religions, cooperates with two further, apparently distinct principles in fashioning the universe: the Forms he uses as patterns, and the receptacle in which

the world is born. As Plato puts it, if the Demiurge is the father of the universe, then the receptacle is its mother.

The *Timaeus* is a dialogue, of course, so in addition to these cosmic characters there is also a small cast of dramatic characters. These consist of Socrates and three others. In fact, Socrates counts the others in the very first sentence of the dialogue, which literally begins: "One, two, three." Plato is, as he sometimes likes to do, alluding to a major theme of a dialogue in its opening words. As we'll see, this dialogue is going to have a great deal to say about mathematics, and Plato marks this by putting numbers quite literally at the front of the *Timaeus*. By the way, the best example of this Platonic trick is the *Republic*, which begins with Socrates saying: "I went down to the Piraeus," that is, to the port near Athens. It can't be unintentional that the dialogue with the cave allegory in it begins with someone saying "I went down."

When our action begins in the *Timaeus*, it turns out that the *Republic* is very much on our minds. Socrates begins the dialogue by summarizing a discussion he and his colleagues have had on the previous day (17a–19b). What he says sounds suspiciously like the content of the *Republic*. So this little group has already heard a depiction of the ideal city. This dialogue will fill out the picture, but will paint on an even larger canvas. In the *Republic* we had an extended parallel between the soul and the city. Now a further parallel is drawn between the city and the universe. We will be shown the ideal order that reigns in the universe as a whole, and not just in one city. First, though, we are shown a picture of the ideal city in action. Socrates' associate Critias—our friend from the *Charmides*, if you remember him—tells a story that the great Athenian statesman Solon supposedly heard in Egypt (21a–25d). Once upon a time, Solon was told, Athens faced down an invasion from a mighty foreign land. No, it's not the story about Persia again. This ancient invasion came from a land which lay in the other direction from Persia, out in the Atlantic Ocean, beyond the Straits of Gibraltar—the continent of Atlantis. This myth is taken up again in a second dialogue, the *Critias*, a companion-piece to the *Timaeus* which is very rarely read nowadays, never mind in the Middle Ages.

After this prefatory material we arrive at the account of Timaeus. He is introduced as a philosopher and astronomer, and once he starts talking he is hardly interrupted. In fact this Platonic dialogue consists mostly of a monologue, delivered by Timaeus. One might suspect that Plato has gotten tired of dialogues and decided to sit down and write a didactic theoretical treatise for a change. He's finally going to stop playing dramatic games and tell it like it is. But that impression is undermined to some extent by Timaeus' own description of his speech. Because of the very nature of the topic, he announces that he will only be able to present, as he puts it, a "likely

story” or “plausible account” (*eikos logos*, 29c). The reason for his limited ambition is that he will be dealing with the physical cosmos. It is, he says, a realm of change and becoming, and thus is susceptible only to opinion. Certainty and knowledge apply only to the world of true being: the Forms. As I said at the end of the previous chapter, old-fashioned Platonism seems to be alive and well in the *Timaeus*.

Something else that is alive and well is the physical universe itself. Timaeus begins his account by saying that the Demiurge, being good, wanted to create the best universe possible. He thus decided that the universe should itself be a living being with a soul, and should imitate insofar as is possible the Form of living being (30b–d). To put it another way, the universe is an animal that is designed as a copy of the Form of Animal.¹ Timaeus assumes that the universe is a sphere, because the sphere is the most perfect shape (33b). He says it is made of the four elements we know from Empedocles: fire, air, water, and earth. He gives an argument for this: if the universe is to be visible there must be fire, and if it is to be tangible there must be earth (31b). But the universe would not be perfectly bonded together if it did not contain proportions between these two extremes, fire and earth (31c–32a). So for the sake of mathematical completeness, the Demiurge includes air and water in between these first two elements.

Plato is here assuming something that will become familiar in Aristotle and most other ancient thinkers: the four elements mix together but tend towards being arranged in layers, with heavy, solid earth settling at the midpoint of the spherical universe, light, subtle fire dominating at the periphery, and water and air in between. What we now think of as the planet Earth is located at the center of the universe, with the heavens rotating around it. All of this, of course, is in keeping with everyday experience—which is just what we would expect from Timaeus’ “likely story.” We see that flame flickers upwards in air, that clods of earth and stones sink in water and fall down through air. As for the heavenly bodies, they cannot be observed up close, but Timaeus says that they too are compounded out of the elements (32c). He adds that their regular motions create time, like a cosmic clock (37d–38e). The heavens are the most divine parts of the universe, but the universe as a whole is divine, a god made by the greater god who is the Demiurge (34b).

Timaeus’ reasoning throughout this section may itself strike us as an odd mixture that moves in different directions. On the one hand, he appeals to empirical observation—as I said, the ideas about the elements are supposed to explain what we see around us. On the other hand, he invokes mathematical symbolism such as we might expect from a Pythagorean. One example is his idea that air and water must exist in order that there be some proportion bonding together fire and earth. The mathematics becomes even more dominant as Timaeus goes along, particularly

in a complicated passage about the creation of soul by the Demiurge. This is explained by means of a detailed analysis of geometrical ratios, with the Demiurge seeking to create an ideally proportionate mixture which will constitute the soul (35b–36b). The Demiurge then weaves this well-proportioned soul with the body of the cosmos, to create a living sphere—as perfect an animal as can exist in the physical realm. This animal is the universe in which we live.

But how do we come to live in the universe? Did the Demiurge create us? No, he did what all good executives do: he delegated. In a characteristically inscrutable little passage, Plato has Timaeus endorse the ancient myths about the generation of the gods, more or less as we know them from Homer and Hesiod. We should, he says, simply take at face value the accounts of the gods that have been handed down to us, even without proof (40d–e). Thus we can go ahead and believe that there is a family of gods on hand to help the Demiurge, who include Kronos, Zeus, Hera, and so on. Unlike the Demiurge, these gods are actually born from one another, just as Hesiod says (40e–41a). If this passage could be read by Xenophanes, the Pre-Socratic who reveled in criticizing the epic poets, we might imagine him sighing in frustration here. Alternatively, he might indulge in a conspiratorial smile, if he assumed that Plato is not serious, and that this ratification of the traditional Greek gods is laced with irony. It's hard to know which would be the right reaction: as usual, Plato is keeping his cards close to his chest.

It is, in any event, these lesser gods who do the dirty work of fashioning the human body. The Demiurge takes care of separating out portions of soul, but his helper gods design our bodies and produce them out of the four elements (42e–43a), making our heads roughly spherical in imitation of the sphere of the universe (44d). The only reason we need our bodies from the neck down, as it turns out, is so that our heads don't roll around and get stuck in ditches. (Again, you have to wonder whether Plato is kidding, but the whole account is delivered with an apparently straight face.) Timaeus also explains how human eyesight works, and at some length. Without getting into these details too much, Plato upholds the view that our vision goes out to the objects we see: the eyes emit rays, which coalesce with light to make seeing happen (45b–d). Thus vision is comparable to using a stick to reach out and touch the visual objects. Yet again, Plato's account will strike us as rather strange. But by assimilating vision to a kind of touch, it explains how we can interact with things at a distance by seeing them. The stipulation that the visual ray must coalesce with light further explains why we can't see in the dark. So this visual ray theory has its advantages, and after Plato it was one of the dominant ways of understanding vision for many centuries.² At any rate, Timaeus praises eyesight as the most important of the senses, because it allows us to see the regular motions of

the heavenly bodies—we should aspire to imitate these revolutions in the motions of our own souls (47a).

This, then, is Timaeus' description of how the Demiurge and his helper gods fashion the universe and the humans living within it. But Timaeus isn't done. He has so far left out an important part of the story: the receptacle (48e–52d). The receptacle is none of the objects that surround us, but is rather what contains those things. We cannot observe it directly. We know it exists only because there must be some matrix or spatial arena in which things move and change into one another. And things certainly do change; for instance, when water evaporates and turns into air. Not for the first time, Plato sounds a bit like a Heraclitean here, as he has Timaeus describe the physical world as a realm of constant change and flux. Without the receptacle there would be no stability at all in the region here around us, below the orderly and unchanging heavens. The receptacle itself is unchanging, but featureless—Timaeus compares it to the odorless fluid to which people add further smells when they are making perfume (50e). So it can only be grasped indirectly, by what Timaeus calls “bastard reasoning.”

Thus Plato has added a third rung to his metaphysical ladder. In dialogues like the *Phaedo* and the *Republic* we were acquainted with two kinds of metaphysical items: the unchanging Forms and the changing things that participate in them. Now we are given what Timaeus calls “a third kind,” namely the receptacle in which these participating things can exist. Timaeus brings in the receptacle in order to illustrate how the universe is the product of not only divine intellect but also what he calls “necessity” (*anangke*) (47e–48a). The Demiurge and helper gods are constrained in certain ways as they fashion the universe. For instance, objects in the universe must take up space, since they are made in the spatial realm of the receptacle. This means that they can collide and interfere with one another. There are other, more subtle limitations too. Timaeus gives the example of the human skull, which turns out to be a kind of compromise. If it were thinner, our brains could be larger and we would be more intelligent, but if it were thicker it would provide better protection (75a–c). Its present thickness is thus the product of necessity: the gods do the best they can with the materials that are physically possible.

The example of the skull comes quite a bit later in the dialogue, towards the end of Timaeus' attempt to analyze the physical world, taking into account not only the Demiurge's wisdom but also the constraints imposed by necessity. As this part of the likely account takes shape, it's one shape in particular that dominates the discussion: triangles. Of course Plato loved geometry—remember the legend that a sign on the Academy said only those who had studied geometry should enter. And his passion for the subject is never more evident than here in the *Timaeus*. We are

told that the four elements are not strictly speaking elements, in the sense of being the most basic constituents of bodies. Instead, just as our bodies are made up of fire, air, water, and earth, so those elemental bodies are made of triangles (53c–d). To be precise, the four elements are made out of two kinds of triangles. Earth is made of one kind (55d–e): triangles with one right angle and two further equal angles (we'd say that the angles are 90, 45, and 45 degrees). If you fuse two such triangles, you get a square, and if you fuse six squares along their edges, you get a cube. This explains the solidity of earth: at what we might call the molecular level, it is made of microscopic cubes that are packed together. All the other elements are made of little triangles, each of which is half of an equilateral triangle. Once they join as equilateral triangles, these can in turn be combined to form the surfaces of various three-dimensional shapes (56a–c).

If you were paying attention in geometry class when you were a kid, you will know that these shapes are called polyhedra. And if you've ever wondered why five such polyhedra, the cube, pyramid, octahedron, dodecahedron, and icosahedron, are called the "Platonic solids," then now you know: these are the five solids mentioned in this part of the *Timaeus*.³ Next, Plato invokes the geometrical features of the solids to explain physical phenomena. We already saw that the solidity of cubes explains the solidity of earth. Likewise, fire consists of tiny pyramids, and the sharp points of these pyramids accounts for the cutting and destructive nature of fire. Another advantage of this account is in explaining how fire, air, and water turn into one another: the three-dimensional molecules are broken up into their triangular atoms, and these triangles then re-form, so that one Platonic solid arises from another (56e–57a). Incidentally, earth, being made of that other kind of triangle, cannot change into the other elements (56d)—a view which Aristotle will go on to reject.

Taking off from this likely account of the elements in terms of geometrical atomism, Timaeus goes on to discuss why some things are hard and others soft, some heavy and others light, some rough and others smooth; why things have color and taste; and why the human body is put together the way it is. In one significant passage he assigns the three parts of the soul familiar from the *Republic* to parts of the body (69c–71e). Reason is located in the brain, spirit in the heart, and appetite in the liver. This too is something that Aristotle will go on to reject. Indeed, a debate will rage for centuries as to whether the rational and perceiving part of our soul is associated with the brain or the heart, with Plato being the standard-bearer for the brain and Aristotle and the Stoics insisting that the heart is, in every sense, central. Only in the second century AD would the doctor Galen put this debate to rest. He performed anatomical experiments—often as a public display—which proved that

animal motion is controlled via nerves that stem from the brain. One powerful demonstration was to snip the nerves in the neck of a live animal, rendering the animal lame.

But let's step back from these anatomical details and look one last time at the bigger picture: the cosmology of the *Timaeus*. Plato's cosmic recipe might remind us of the Pre-Socratics. The ingredients of his physical universe include four elements (borrowed from Empedocles), and a divine mind (that's a healthy portion of Anaxagoras), who produces a world in constant change (a dash of Heraclitus), which is fundamentally mathematical (a full measure of Pythagoreanism). But as always, it would be a mistake to reduce Plato to his sources. The *Timaeus* provides us with an ambitious and novel account of the universe and its making. It shows a deep commitment to the idea that the universe is providentially ordered. Though Plato doesn't use any phrase like "the best of all possible worlds," he does stress that the Demiurge is not envious, and wants to make this world as good and beautiful as possible. On the other hand, there are limitations on that possibility. The triangles out of which the elements arise are, Timaeus tells us, the most beautiful and perfect shapes (54a). But the very fact that the elements are made of shapes has to do with the nature of the receptacle, and in general the Demiurge's choices are limited by physical necessity.

This is not the untrammelled creative activity of the Christian God worshipped by medieval monks. Rather, the universe is the work of a divine craftsman who has no choice about which blueprints to use, and who is restricted as regards his materials. Another, equally profound difference is this: the God of Christianity, Judaism, and Islam is a personal deity. He rewards and punishes, He grows angry, He loves his creatures. The Demiurge does none of these things. He may be providential and even generous, but in making the universe his attitude is more like aesthetic taste than love. Indeed, there is a good Platonic reason why the Demiurge cannot love us. The Demiurge is divine and perfect, and needs nothing. And in other dialogues, Plato shows us that love is always bound up with need, with unfulfilled desire, with a longing for a beauty which we strive to possess.